Case Study 31: String-Based Examination Answer Evaluation

1. Problem Statement

Compare student answer strings against a standard answer key character-by-character, calculate correct, incorrect, and unanswered counts, apply negative marking rules, and generate student-wise examination scores.

2. Objectives

- Compare multi-choice response strings against an answer key sequence using basic loops.
- Incorporate positive points, negative deduction penalties, and zero scoring for unattempted items.
- Output student evaluation metrics including total score, accuracy percentage, and final outcome.

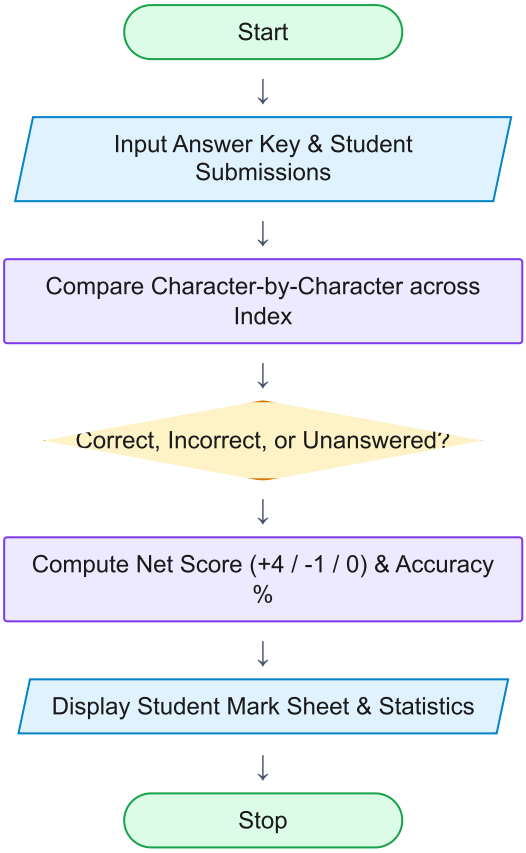
3. Assumptions

- Correct Answer reward = **+4 marks**.
- Incorrect Answer penalty = **-1 mark**.
- Unanswered character (represented by `'X'` or `'?'`) = **0 marks**.
- Total test length = 10 questions (Maximum score = 40 marks).

4. Algorithm

1. Start.
2. Read standard answer key string and student answer records `[ (roll_no, name, answer_string), ... ]`.
3. For each student record:
   - Initialize `correct = 0`, `incorrect = 0`, `unanswered = 0`.
   - Loop through index `i` from 0 to `len(key) - 1`:
     - If `ans_str[i] == 'X'`, increment `unanswered`.
     - Else if `ans_str[i] == key[i]`, increment `correct`.
     - Else increment `incorrect`.
   - Calculate score: `total_score = (correct * 4) - (incorrect * 1)`.
   - Calculate accuracy percentage: `accuracy = (correct / (correct + incorrect)) * 100` (if attempted > 0).
4. Display student marks report and summary performance breakdown.
5. Stop.

5. Flowchart



6. Pseudocode

```
START INPUT key_string INPUT student_records FOR EACH student IN
student_records DO roll = student.roll name = student.name answers =
student.answers correct = 0 incorrect = 0 unanswered = 0 FOR i FROM 0 TO
length(key_string) - 1 DO char_key = key_string[i] char_ans = answers[i] IF
char_ans == "X" THEN unanswered = unanswered + 1 ELSE IF char_ans == char_key THEN
correct = correct + 1 ELSE incorrect = incorrect + 1 ENDIF ENDFOR score = (correct
* 4) - (incorrect * 1) attempted = correct + incorrect IF attempted > 0 THEN
accuracy = (correct / attempted) * 100.0 ELSE accuracy = 0.0 ENDIF PRINT roll,
name, correct, incorrect, unanswered, score, accuracy ENDFOR STOP
```

7. Complete Python Source Code

```
     # String-Based Examination Answer Evaluation def evaluate_exam(answer_key,
student_submissions): print("=" * 72) print(" EXAMINATION ANSWER EVALUATION
REPORT") print("=" * 72) print(f"Answer Key: {answer_key}") print("-" * 72)
print(f"{'Roll No':<10}{'Name':<14}{'Correct':<10}{'Wrong':<8}{'Skip':<8}{'Score':
<10}{'Accuracy':<10}") print("-" * 72) for roll, name, ans in student_submissions:
correct = 0 incorrect = 0 unanswered = 0 for i in range(len(answer_key)): if ans[i]
== 'X': unanswered += 1 elif ans[i] == answer_key[i]: correct += 1 else: incorrect
+= 1 score = (correct * 4) - (incorrect * 1) attempted = correct + incorrect
accuracy = (correct / attempted * 100.0) if attempted > 0 else 0.0 print(f"{roll:
<10}{name:<14}{correct:<10}{incorrect:<8}{unanswered:<8}{score:<10}{accuracy:
<10.1f}%") print("=" * 72) # Master Key (10 Questions) master_key = "ABCDABCDAB" #
Submissions: (Roll, Name, Answer String where 'X' means unattempted) submissions =
[ ("R101", "Aditya Rao", "ABCDABCDAB"), # Perfect score ("R102", "Bhavya Garg",
"ABCDABCCAX"), # 7 correct, 2 wrong, 1 skip ("R103", "Charan Tej", "XXXXXABCDA"), #
5 skips, 4 correct, 1 wrong ("R104", "Deepa Nair", "DCBADCBADC") # All incorrect ]
evaluate_exam(master_key, submissions)
```

8. Test Case and Expected Output

Evaluation Output for Test Batch:

```
     ========================================================================
EXAMINATION ANSWER EVALUATION REPORT
======================================================================== Answer
Key: ABCDABCDAB ------------------------------------------------------------------
----- Roll No Name Correct Wrong Skip Score Accuracy ----------------------------
---------------------------------------- R101 Aditya Rao 10 0 0 40 100.0% R102
Bhavya Garg 7 2 1 26 77.8% R103 Charan Tej 4 1 5 15 80.0% R104 Deepa Nair 0 10 0
-10 0.0% ========================================================================
```

9. Conclusion

The String-Based Examination Answer Evaluation system efficiently processes test submissions, applying positive rewards and negative penalization to compute scores and accuracy rates accurately.